# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning**

Shaping the Future of Artificial Intelligence

## Minute to Minute Programme

**GUEST OF HONOUR**

M Balasubramaniam

CEO STRATINFINITY INC
DEPUTY CHAIRMAN MOE INVESTOR COMMITTEE
FOUNDER CHAIRMAN IIIT SRICITY

**CHIEF GUEST**

Dr. Gopichand Katragadda

CEO, MYELIN FOUNDRY

**CHIEF PATRON**

Prof. Bhartendu K. Singh

DIRECTOR, IIITDM JABALPUR

**SPECIAL INVITEES**

Prof. Aparajita Ojha
DISTINGUISHED FACULTY, IIITDM JABALPUR

Prof. Pritee Khanna
SENIOR FACULTY & DEAN (RSPC), IIITDM JABALPUR

Prof. Atul Gupta
SENIOR FACULTY, IIITDM JABALPUR

**CONVENER**

Dr. Rakesh K Sanodiya
IIITDM JABALPUR

**CO-CONVENER**

Dr. Ranjeet Kumar Ranjan
IIITDM JABALPUR

## Welcome to Deep Learning Summer School!

### Kickstart Your AI Journey

Dive deep into neural networks and advanced AI concepts this summer.

### Join Us for Knowledge

Engage in hands-on learning and expert-led sessions tailored for deep learning enthusiasts.

# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning: Shaping the Future of Artificial Intelligence**

## Inauguration Session

**19-05-2025**

| Session No. | Activity | Timings |
|---|---|---|
| 1 | Arrival of Guests | 9:30 AM |
| 2 | Introduction by Master of Ceremony | 9:30 am- 9:32 am |
| 3 | Felicitation of Guests | 9:32 am - 9:34 am |
| 4 | Welcome Address by Dr. Rakesh Kumar Sanodiya | 9:35 am - 9 :40 am |
| 5 | Speech by Special Invitee Prof. Aparajita Ojha | 9:40 am - 9:45 am |
| 6 | Speech by Special Invitee Prof. Pritee Khanna | 9:45 am - 9: 50 am |
| 7 | Speech by Special Invitee Prof. Atul Gupta | 9:50 am - 9:55 am |
| 8 | Inaugural Address by Chief Patron Prof. Bhartendu K. Singh | 10:00am - 10:05 am |
| 9 | Address by Chief Guest Dr. Gopichand Katragadda | 10:05 am - 10:15 am |
| 10 | Address by Guest of Honour M Balasubramaniam | 10:15 am - 10:27am |
| 11 | Vote of Thanks by Dr. Ranjeet Kumar Ranjan | 10:27 am - 10:30 am |

**PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur**
**Dumna Airport Road, Jabalpur 482005, Madhya Pradesh, India**
**Email : rakesh.s@iiitdmj.ac.in Phone no. : 8770120278**

# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning: Shaping the Future of Artificial Intelligence**

## Week - 1

## Foundations of Deep Learning

***Objective*** : Build a strong mathematical and practical foundation in deep learning

| Day | Date | Theory<br>(2 hours:11:00 am to 01:00PM) | Practical<br>(2 hours:02:30 am to 4:30PM) | Session-Chairs |
|---|---|---|---|---|
| 1 | 19-05-2025 | Introduction<br>to AI, Machine Learning and Deep Learning<br><br>By Prof. Aparajita Ojha, PDPM IIITDM Jabalpur | Setting<br>Up the Environment<br>-Maths<br>By Dr Rakesh K Sanodiya, PDPM IIITDM Jabalpur + 3 | Gaurav Bhardwaj |
| 2 | 20-05-2025 | **(2 HOURS:09:00 AM TO 11:00PM)**<br>Neural<br>Networks Basics<br>- Perceptrons, activation functions<br>- Forward and backward propagation<br>By Prof. Sumantra Dutta Roy, IIT Delhi<br><br>**(2 HOURS:11:00 AM TO 01:00PM)**<br>Mathematics<br>for Deep Learning<br>- Linear Algebra, Calculus, and Probability<br>By Prof. Sumantra Dutta Roy, IIT Delhi | Python<br>for Deep Learning<br>- NumPy<br>- Introduction to Autograd (Automatic Differentiation)<br><br>By Dr. Rakesh Kumar Sanodiya IIITDMJ | Ishita Tyagi |
| 3 | 21-05-2025 | | Building<br>a Simple Neural Network<br>- Implementing a perceptron from scratch<br><br>By Dr. Samir Jain, alfaTKG | Sunaina Singh |
| 4 | 22-05-2025 | Training<br>Neural Networks<br>- Loss functions and gradient descent<br>- Optimization algorithms (SGD, Adam, RMSProp)<br><br>By Dr. Shiv Ram Dubey, IIIT Allahabad | Optimizing<br>Neural Networks<br>- Implementing gradient descent<br>- Hyperparameter tuning (learning rate, batch size effects)<br>- Visualizing loss curves<br>By Dr. Samir Jain, alfaTKG | Supriya Sharma |
| 5 | 23-05-2025 | Deep<br>Learning Frameworks<br>- TensorFlow vs. PyTorch<br>- Overview of Keras<br><br>By Dr. Chandra Sekhar Vorugunti, Samsung | Hands-on<br>with PyTorch/Keras<br>- Building and training a simple model<br><br>By Dr. Chandra Sekhar Vorugunti | Gaurav Bhardwaj |

**PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur**
**Dumna Airport Road, Jabalpur 482005, Madhya Pradesh, India**
**Email : rakesh.s@iiitdmj.ac.in Phone no. : 8770120278**

# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning: Shaping the Future of Artificial Intelligence**

## Week - 2

## Core Deep Learning Architectures

***Objective*** : **Understand and implement key deep learning architectures.**

| Day | Date | Theory (2 hours:11:00 am to 01PM) | Practical (2 hours:02:30 am to 4:30PM) | Session-Chairs |
|---|---|---|---|---|
| 6 | 26-05-2025 | Convolutional Neural Networks (CNNs)<br>- Basics of CNNs (convolution, pooling, filters)<br>By Dr. Shivram Dubey, IIITA | Building a CNN<br>- Image classification using CIFAR-10<br><br>By Dr. Poornima Singh Thakur, ABV IIITM Gwalior | Gaurav Bhardwaj |
| 7 | 27-05-2025 | Advanced CNNs<br>- Transfer learning and pre-trained models (ResNet, VGG)<br>By Dr. Rama Krishna Gorthi, IIT Tirupati | Transfer Learning with CNNs<br>- Fine-tuning a pre-trained model<br><br>Dr Rakesh K Sanodiya, PDPM IIITDM Jabalpur | Ishita Tyagi |
| 8 | 28-05-2025 | Recurrent Neural Networks (RNNs) and its Applications in NLP<br>- Basics of RNNs, LSTMs, and GRUs<br>By Dr. Arijit Mondal IIT Patna | Building an RNN<br>- Text generation or time-series prediction<br>By Mr. Fazail Amin / Mr Sourabh Dattatrya | Sunaina Singh |
| 9 | 29-05-2025 | Self-attention and Transformer models<br>By Miss. Niharika Ahuja, GOOGLE | Implementing Transformers<br>- Text summarization or translation<br><br>By Miss. Niharika Ahuja, GOOGLE | Supriya Sharma |
| 10 | 30-05-2025 | Unsupervised Learning, and Generative Modeling<br>- Autoencoders, Variational Autoencoder<br><br>By Dr. Jignesh S. Bhatt, IIITV | Building an autoencoder for (1) representation learning (2) Image denosing<br><br>By , Dr. Jignesh S. Bhatt, IIITV | Ishita Tyagi |
| 11 | 31-05-2025 | Large Language Models (LLMs)<br>- Transformer architectures (GPT, BERT, LLaMA)<br>- RLHF (Reinforcement Learning from Human Feedback)<br><br>Mr. Rushendra Sidibomma, ML Research Engineer, AutomationEdge Technologies, | Hands-on with LLMs<br>- Fine-tuning a pre-trained LLM<br>- Experimenting with prompt engineering<br><br>Mr. Rushendra Sidibomma, ML Research Engineer, AutomationEdge Technologies, | Sunaina Singh |

**PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur**
**Dumna Airport Road, Jabalpur 482005, Madhya Pradesh, India**
**Email : rakesh.s@iiitdmj.ac.in  Phone no. :  8770120278**

# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning: Shaping the Future of Artificial Intelligence**

## Week - 3

## Advanced Topics and Cutting-Edge Research

***Objective*** : Explore advanced and emerging trends in deep learning.

| Day | Date | Theory (2 hours:11:00 am to 01PM) | Practical<br>(2 hours:02:30 am to 4:30PM) | Session-Chairs |
|---|---|---|---|---|
| 12 | 02-06-2025 | Introduction to Generative Adversarial Network,<br>By Prof. R Bala Subramanyam, IIT Roorkee | Training a GAN for Data augmentationBy Ms Shubhangi Chaturvedi, ABV IIITM Gwalior/Dr. Rakesh Kumar Sanodiya, IIITDMJ | Gaurav Bhardwaj |
| 13 | 03-06-2025 | Diffusion Models<br>- Introduction to diffusion models<br>- Denoising Diffusion Probabilistic Models (DDPMs)<br>- Text-to-image models<br><br>By Prof. Aparajita Ojha | Hands-on Diffusion Models<br>- Implementing Stable Diffusion for image generation<br>By Dr. Chandra Sekhar, Samsung | Ishita Tyagi |
| 14 | 05-06-2025 | Explainable AI (XAI)<br>- Interpretability in deep learning<br>- Techniques for model explainability<br>By Prof RKonda Reddy, IIT Hyderabad | Visualizing Model Decisions<br>- Using tools like SHAP or LIME<br><br>By Prof RKonda Reddy, IIT Hyderabad | Supriya Sharma |
| 15 | 06-06-2025 | Reinforcement Learning (RL)<br>- Basics of RL and Deep Q-Learning<br>- Introduction to Multi-Agent RL<br><br>Dr. Madan Mohan Vasu | Hands-on RL<br>- Building a simple RL agent<br>Dr. Madan Mohan Vasu | Sunaina Singh |

**PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur**
**Dumna Airport Road, Jabalpur 482005, Madhya Pradesh, India**
**Email : rakesh.s@iiitdmj.ac.in  Phone no. :  8770120278**

# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning: Shaping the Future of Artificial Intelligence**

## Week - 4

## Capstone Projects & Real-World Applications

***Objective*** : Explore advanced and emerging trends in deep learning.

| Day | Date | Theory (2 hours:11:00 am to 01PM) | Practical (2 hours:02:30 am to 4:30PM) | Session-Chairs |
|---|---|---|---|---|
| 16 | 09-06-2025 | Capstone Project Kickoff<br>- Participants select project themes (Computer Vision, NLP, Healthcare, Remote Sensing)<br><br>By Prof. Pritee Khanna, and Prof. Aparajita Ojha IIITDMJ | Project Ideation & Dataset Selection<br>- Defining the problem statement<br>- Choosing datasets and evaluation metrics<br><br>Dr.Ravi Ranjan Karan, Accenture | Gaurav Bhardwaj |
| 17 | 10-06-2025 | Computer Vision Applications<br>- Image segmentation and classification<br>By Mr.PRANTIK SEN. Amazon, Dr. Rakesh Kumar Sanodiya | AI in Remote Sensing<br>- Using Domain Adaptation and Transfer Learning models<br>By Dr. Anish Chand Turlapat, IIIT Sri City | Ishita Tyagi |
| 18 | 11-06-2025 | AI in Healthcare<br>- Medical imaging and predictive analytics<br><br>By Dr. subramanya gorthi, IIT Tirupati (12 to 1), Prof. Tanveer, IIT Indore (11am to 12pm) | Smart Agriculture<br>By Dr. Sudeep Rai IIT Roorkee,<br>By Dr. Akshay Pandey, IIITDMJ | Sunaina Singh |
| 19 | 12-06-2025 | NLP Applications<br>- Chatbots and machine translation<br>By Dr. Sriparna Saha, IIT Patna | Security Applications by Dr Emmanuel Pilli, MNIT Jaipur | Supriya Sharma |
| 20 | 13-06-2025 | Capstone Project Presentations<br>- Industry mentors and researchers provide feedback<br><br>Dr.Ravi Ranjan Karn, Accenture | Peer and Mentor Feedback<br>- Evaluating projects and discussion | Supriya Sharma |

**PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur**
**Dumna Airport Road, Jabalpur 482005, Madhya Pradesh, India**
**Email : rakesh.s@iiitdmj.ac.in  Phone no. :  8770120278**

# PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

**1st Summer School on Deep Learning**

Shaping the Future of Artificial Intelligence

## Invited Speakers

**Prof. Sumantra Dutta Roy**
IIT DELHI

**Prof. M. Tanveer**
IIT INDORE

**Prof. Aparajita Ojha**
IIITDMJ

**Prof. Pritee Khanna**
IIITDMJ

**Dr. Sriparna Saha**
IIT PATNA

**Dr. Rama Krishna Gorthi**
IIT TIRUPATI

**Dr. Namita Mittal**
MNIT JAIPUR

**Dr. Sandeep Sengar**
CARDIFF METROPOLITAN UNIVERSITY, UK

**Dr. Hemant Aggarwal**
GE HEALTHCARE

**Mr. Chinmay Sharma**
LINKEDIN

**Dr. Arijit Mondal**
IIT PATNA

**Miss. Niharika Ahuja**
GOOGLE

**Dr. Debanjan Konar**
SAMSUNG

**Dr. Ravi Ranjan Karn**
ACCENTURE

**Dr. Shiv Ram Dubey**
IIIT ALLAHABAD

**Mr. Jaswin Kasi**
AMAZON

**Mr. Prantik Sen**
AMAZON

**Mr. Nirakar Padhay**
COLLINSON

**Dr. Chandra Sekhar**
SAMSUNG

**Dr. Anish Chand Turlapaty**
IIITS

**Dr. Jignesh S. Bhatt**
IIITV

**Dr. Rakesh Kumar Sanodiya**
IIITDMJ

**Dr. Samir Jain**
ALFATKG

**Dr. Poornima Singh Thakur**
ABV IIITM GWALIOR

**Mr. Rushendra Sidibomma**
AUTOMATIONEDGE TECHNOLOGIES

**Prof. R Bala Subramanyam**
IIT ROORKEE

**Prof R Konda Reddy**
IIT HYDERABAD

**Dr. Sudeep Rai**
IIT ROORKEE

**By Dr. Akshay Pandey**
IIITDMJ

**Dr Emmanuel Pilli**
MNIT JAIPUR

**Dr. Matloob Khushi**
BRUNEL UNIVERSITY LONDON

**Dr. Ranjeet Kumar Ranjan**
IIITDMJ

**PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur**
**Dumna Airport Road, Jabalpur 482005, Madhya Pradesh, India**
**Email : rakesh.s@iiitdmj.ac.in  Phone no. : 8770120278**